Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 4: Approved Solutions

Implications, Vacuous Truth, and Negation

Covers Chartrand & Zhang, Section 1.3, pp. 25–33

Status of this set

Ungraded practice. Solutions are released with it. Attempt every problem before reading any solution. Fill truth tables in on this page; everything else goes on your own paper.

Part I: How to Practice This Material

One row does all the work

Everything in this section follows from a single fact: $P \Rightarrow Q$ is false in exactly one row, when P is true and Q is false. Every other result is a consequence.

Three things to have automatic

1. $P \Rightarrow Q \equiv (\sim P) \vee Q$ (Thm 1.48). When an implication is awkward, rewrite it this way and use the laws you already have.
2. $\sim(P \Rightarrow Q) \equiv P \wedge (\sim Q)$ (Thm 1.50). The negation of an implication is a *conjunction*, never another implication.
3. **A false hypothesis makes the implication true.** Check the hypothesis first — it often settles the question with no further work.

Three mistakes to expect

1. **Negating an implication into an implication.** The negation of "if P then Q" is "P and not Q."
2. **Insisting a true implication needs a real connection.** It does not. The definition looks only at truth values.
3. **Calling a vacuously true statement false or meaningless.** If nothing satisfies the hypothesis, the implication is true.

Part II: Practice Problems

Section A — Truth values

1. Determine whether each implication is true or false, and give a one-line reason.

(a) If 9 is a perfect square, then 9 is odd.

(b) If 9 is a perfect square, then 9 is even.

(c) If 8 is a perfect square, then 8 is odd.

(d) If $\pi > 4$, then $\pi > 3$.

Approved solution

(a) True. Hypothesis true, conclusion true — row 1.
(b) False. Hypothesis true ($9 = 3^2$), conclusion false (9 is odd). This is the only row that gives a false implication.
(c) True. 8 is not a perfect square, so the hypothesis is false and the implication is true — regardless of anything about 8 being odd.
(d) True. $\pi \approx 3.14$, so the hypothesis $\pi > 4$ is false.

Check the hypothesis first. In (c) and (d) the conclusion never had to be examined. Students who evaluate both sides every time are doing avoidable work.

2. Construct a truth table for $(P \Rightarrow Q) \wedge (\sim Q)$, then state in one sentence when it is true.

P	Q	$P \Rightarrow Q$	$\sim Q$	$(P \Rightarrow Q) \wedge (\sim Q)$
T	T	T	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Approved solution

True in exactly one row — when P and Q are both false.
Read that as a sentence: *if the implication holds and the conclusion fails, then the hypothesis must fail too.* That is the reasoning pattern behind proof by contrapositive, which arrives on Wednesday and again in Chapter 3.

Section B — Vacuous truth

3. Decide whether each statement is true, and say why.

(a) Every integer n with $n^2 < 0$ is prime.

(b) Every element of the empty list is greater than 100.

(c) Every prime number greater than 2 is odd.

Approved solution

(a) True, vacuously. No integer has $n^2 < 0$, so nothing can serve as a counterexample.
(b) True, vacuously. The empty list has no elements to violate the condition. In Python, `all(x > 100 for x in [])` returns `True` for exactly this reason.
(c) True — but *not* vacuously. Here the hypothesis is satisfied by infinitely many numbers, and the claim genuinely holds for all of them. This part is included to make sure "vacuously true" does not become a reflex answer for every universal claim.

The distinction: (a) and (b) are true *because nothing satisfies the hypothesis*; (c) is true *despite* many things satisfying it.

Section C — Negating implications

4. Write the negation of each. Phrase it the way a person would say it.

(a) If the connection drops, the transaction is rolled back.

(b) If n is divisible by 6, then n is divisible by 4.

(c) If a student passes the final, then the student passes the course.

(d) If the array is empty, then the function returns zero.

Approved solution

Each negation is a *conjunction*: hypothesis holds, conclusion fails.
(a) The connection drops and the transaction is not rolled back.
(b) n is divisible by 6 and n is not divisible by 4. (This is satisfiable — take $n = 6$, which shows the original implication is false.)
(c) A student passes the final and does not pass the course.
(d) The array is empty and the function does not return zero.

Any answer beginning with "if" is wrong. That is the single check to run on this whole problem.

Section D — Implications with open sentences

5. For a real number x, let $P(x) : x = 3$ and $Q(x) : x^2 = 9$.

(a) State $P(x) \Rightarrow Q(x)$ in words.

(b) Is $P(a) \Rightarrow Q(a)$ true for *every* real number a? Justify.

(c) Is $Q(a) \Rightarrow P(a)$ true for every real a? If not, give a value of a where it fails.

Approved solution

(a) If $x = 3$, then $x^2 = 9$.
(b) Yes. Consider any real a. If $a = 3$ then $a^2 = 9$, so both parts are true and the implication holds. If $a \neq 3$ the hypothesis is false, so the implication holds automatically. Either way it is true.
(c) No. Take $a = -3$: then $Q(-3)$ is true since $(-3)^2 = 9$, but $P(-3)$ is false since $-3 \neq 3$. Hypothesis true, conclusion false — the implication is false.

Note what just happened. $P(x) \Rightarrow Q(x)$ holds everywhere while $Q(x) \Rightarrow P(x)$ does not. Reversing an implication changes it into a genuinely different statement. That reversal has a name, and it is Wednesday's first topic.

Section E — Specifications

6. A system requirement reads:

> *If a request is authenticated, then it is logged.*

(a) Which single situation would show the system violates this requirement?

(b) An auditor finds an unauthenticated request that was logged. Does this violate the requirement?

(c) The system logs nothing at all for a week because it received no requests. Does this violate the requirement?

Approved solution

(a) An authenticated request that was *not* logged. That is the only violation, because it is the only row where the implication is false.
(b) No. The requirement says nothing about unauthenticated requests. Logging one is permitted — it corresponds to row 3, where the implication is true.
(c) No. With no requests at all, the hypothesis is never satisfied and the requirement is vacuously satisfied.

Why this matters in practice. A specification written as an implication constrains exactly one thing. Reading it as though it also forbids the other cases is how requirements get over-implemented — and part (b) is exactly where that mistake shows up.

Section F — Writing

7. In five to eight complete sentences, explain to a classmate why "If $2 + 2 = 5$, then the moon is made of cheese" is a *true* statement. Your explanation should address why this feels wrong and why the definition is set up this way anyway. Use the contract idea from class.

Approved solution

Model answer. The implication is true because its hypothesis, $2+2 = 5$, is false, and an implication is false in only one situation: a true hypothesis with a false conclusion. Since that situation does not arise here, the statement is true. This feels wrong because we expect "if…then" to signal that the hypothesis somehow causes or supports the conclusion, and there is plainly no connection between arithmetic and the composition of the moon. But the truth table is a definition, and it looks only at

truth values. The contract reading makes it easier to accept: an implication is a promise that can be broken only when its condition is met and its consequence fails to follow. Here the condition is never met, so no promise has been broken and the statement stands as true. In practice hypotheses and conclusions are genuinely related, so this oddity almost never causes trouble outside deliberately constructed examples.

Full credit requires both halves: why it is true *and* why the definition is acceptable despite the discomfort. An answer that only recites the truth table earns about half.

Section G — Challenge (optional)

8. Use Theorem 1.48 and the laws from Section 1.2 to show that $P \vee Q \equiv (\sim P) \Rightarrow Q$, without building a truth table.

Approved solution

$$\begin{aligned} (\sim P) \Rightarrow Q &\equiv \big(\sim(\sim P)\big) \vee Q && \text{Theorem 1.48} \\ &\equiv P \vee Q && \text{Double negation (Thm 1.28)} \end{aligned}$$

Two lines. A truth table would also work, but the point of the exercise is that once Theorem 1.48 is available, implications can be folded into the algebra you already know rather than treated as a separate system.

Worth noticing: this says every disjunction is secretly an implication. "It is raining or I brought an umbrella" and "if it is not raining, I brought an umbrella" say exactly the same thing.

Instructor notes

- **Problem 3(c)** exists to stop "vacuously true" becoming a reflex. Expect some students to label it vacuous; the hypothesis there is satisfied infinitely often.
- **Problem 5(c)** sets up Wednesday. Do not resolve the converse here — let the observation sit until Lecture 5 names it.
- **Problem 6(b)** is the best item on the page for CS students. Over-implementing a specification — forbidding what it merely does not require — is a real defect class, and this is its logical skeleton.
- **Problem 4** has a single fast check: any negation starting with "if" is wrong. Say that aloud when returning it.
- Reserved for Homework 3: Section 1.3 Exercises 2, 4, 6, 8, 14 (even-numbered — no printed answers). Exercises 1 and 3 were assigned as self-check because they are odd and students can check themselves against the back of the book.